HOW SAFE IS 5G?

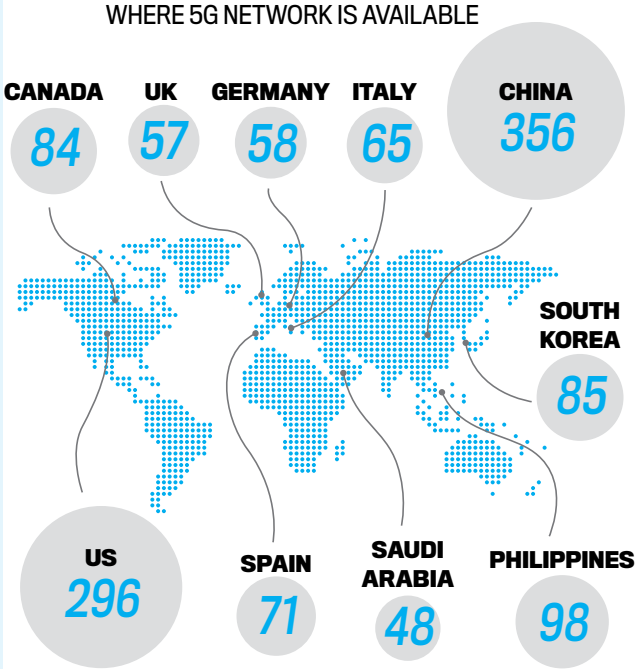
5G uses higher frequencies that are new to mobile phone networks, but are commonly used in other applications such as point-to-point radio links and body scanners for security checks. At these higher frequencies, 5G networks will use a greater number of base stations and of connected objects, which has triggered safety concerns. In January, several airlines cancelled their flights to the US as AT&T and Verizon announced their 5G rollout, fearing that the 5G waves could interfere with instruments such as the altimeters that tell pilots how far they are from the ground.

One worry is that, since 5G is new, there has not been enough time to properly test whether it's safe. According to the World Health Organization (WHO), after conducting much research, no adverse health effect has been linked with exposure to wireless technologies. Health-related conclusions are drawn from studies performed across the entire radio spectrum but, so far, only a few studies have been carried out at the frequencies to be used by 5G, it said. "Tissue heating is the main mechanism of interaction between radio frequency fields and the human body. Radio frequency exposure levels from current technologies result in negligible temperature rise in the human body," says a WHO report.

In June, a single bench of the Delhi High Court dismissed a suit filed by actor Juhi Chawla highlighting what she felt were the adverse effects of 5G. The court imposed a penalty of Rs 20 lakh on her "for wastage of judicial time" but later, a division bench lowered the penalty to Rs 2 lakh and expunged the adverse observations against her.

AROUND THE WORLD

NUMBER OF CITIES
WHERE 5G NETWORK IS AVAILABLE



struct a future-ready network for 5G. It has also tied up with L&T Smart World & Communication for 5G network infrastructure, based on technology expertise from Nokia. Chinese equipment vendors Huawei and ZTE, whose future in India had come under a cloud since border tensions with China in 2020, are reportedly out of the race for the supply of 5G equipment, with telcos preferring to go with other players. Meanwhile, the government-owned Centre for Development of Telematics (C-DoT), in collaboration with private Indian enterprises, is perfecting an indigenous 5G stack (apart from a 4G one), including the core network that BSNL will test before it is offered for commercial sales.

All this requires huge capital investment. Before the auctions, telecom operators had asked for a 90 per cent cut in the reserve price of airwaves from the 2018 levels, arguing that they needed to keep aside a big chunk of funds for setting up infrastructure. But the Centre didn't accede to this demand and reduced the prices for various spectra by an average of 30 to 40 per cent. "The spectrum reserve price is 3.6 times compared to that of Germany, 3.8 times that of the UK and 4.4 times when compared to South Korea. However, India's telecom service revenue is 0.6 times of Germany, 0.7 times of the UK and 0.9 times of South Korea," says Singhal. "That is what creates the imbalance." Paying higher prices for airwaves can impact investments in infrastructure. It will also make it imperative for companies to recoup their expenditure on the 5G rollout at the earliest. The hope for telecom players is that most of the more than 800 million subscribers of the 4G telecom network are expected to make the shift to 5G over a year or so, which will help them earn more revenues out of the data these subscribers use.

There are other challenges that include adding more telecom towers to meet the traffic and also extensive fiberisation of the network so that towers do not get choked. Most sites need fibre-optic connectivity, and the more the amount of fibre, the better. This becomes all the more necessary as 5G matures and India moves into the second and third years of the rollout. As of February, 560,000 kilometres of fibre-optic cable had been laid under the BharatNet policy, according to government data. However, fiberisation has covered only around 30 per cent of the Indian network at present—that needs to increase to 80 per cent for a meaningful launch of 5G, say experts. "The industry and the government are seized of the issue. The Centre's Gati Shakti Sanchar project is an intention to bring down the approvals for laying out fibre and erecting telecom towers," says the industry official.

5G is finally on India's doorstep. It promises a significant change in the way we experience mobile telephony services, both as consumers and from the point of view of enterprises. What now remains is its steadfast implementation, plugging the loopholes in infrastructure and the availability of affordable devices so that the proliferation across the country can take place at a quick pace. ■